Lesson 8.2 Guided Notes

Session 1 · Addition Equations

I can solve an addition equation by subtracting the same amount from both sides. · 6.EE.B.7

Word / Spanish	Meaning — complete the blank	Example
inverse operations / operaciones inversas	Two operations that _____ each other.	+ and −
isolate / aislar	Get the variable _____ on one side.	y = 26
Subtraction Property of Equality / propiedad de sustracción de la igualdad	Subtract the same number from _____ sides. They stay equal.	both sides − 14

Key idea

Subtraction _____ addition.

To solve x + 14 = 40, subtract _____ from both sides so the equation stays _____.

Worked example

Solve 21 + p = 50.

Subtract _____ from both sides.

p = 50 − 21 = _____.

Check: 21 + 29 = _____.

Answer: p = _____

You try

1. Solve each equation.
a. x + 9 = 25 _____
b. 12 + m = 30 _____

_____

2. Solve k + 4.5 = 10. Check your answer.

_____

Sum it up: To solve an addition equation, I ___ because ___.

_____

Session 2 · Subtraction Equations

I can solve a subtraction equation by adding the same amount to both sides. · 6.EE.B.7

Word / Spanish	Meaning — complete the blank	Example
Addition Property of Equality / propiedad de suma de la igualdad	Add the same number to _____ sides. They stay equal.	both sides + 9
difference / diferencia	The answer when you _____.	in m − 9 = 14, it is 14
isolate / aislar	Get the variable _____ on one side.	m = 23

Key idea

Addition _____ subtraction.

To solve p − 7 = 12, add _____ to both sides.

Worked example

Solve m − 9 = 14.

Add _____ to both sides.

m = 14 + 9 = _____.

Check: 23 − 9 = _____.

Answer: m = _____

You try

1. Solve each equation.
a. x − 7 = 19 _____
b. n − 15 = 40 _____

_____

2. Solve p − 2.5 = 4. Check your answer.

_____

Sum it up: To solve a subtraction equation, I ___ because ___.

_____

Session 1 · Model, reason, and explain

Addition Equations · 6.EE.B.7

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Use a balance model to solve x+8=23. Explain why subtracting 8 from only one side would not preserve the equality.

Story quantities	Equal groups / balance	Equation	Substitute to check

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses an unexplained move-across rule or changes one side only.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Subtraction Equations · 6.EE.B.7

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Solve n−7=18 using a balance explanation. Compare its inverse step with that for n+7=18.

Story quantities	Equal groups / balance	Equation	Substitute to check

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: subtracts again to solve a subtraction equation.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?